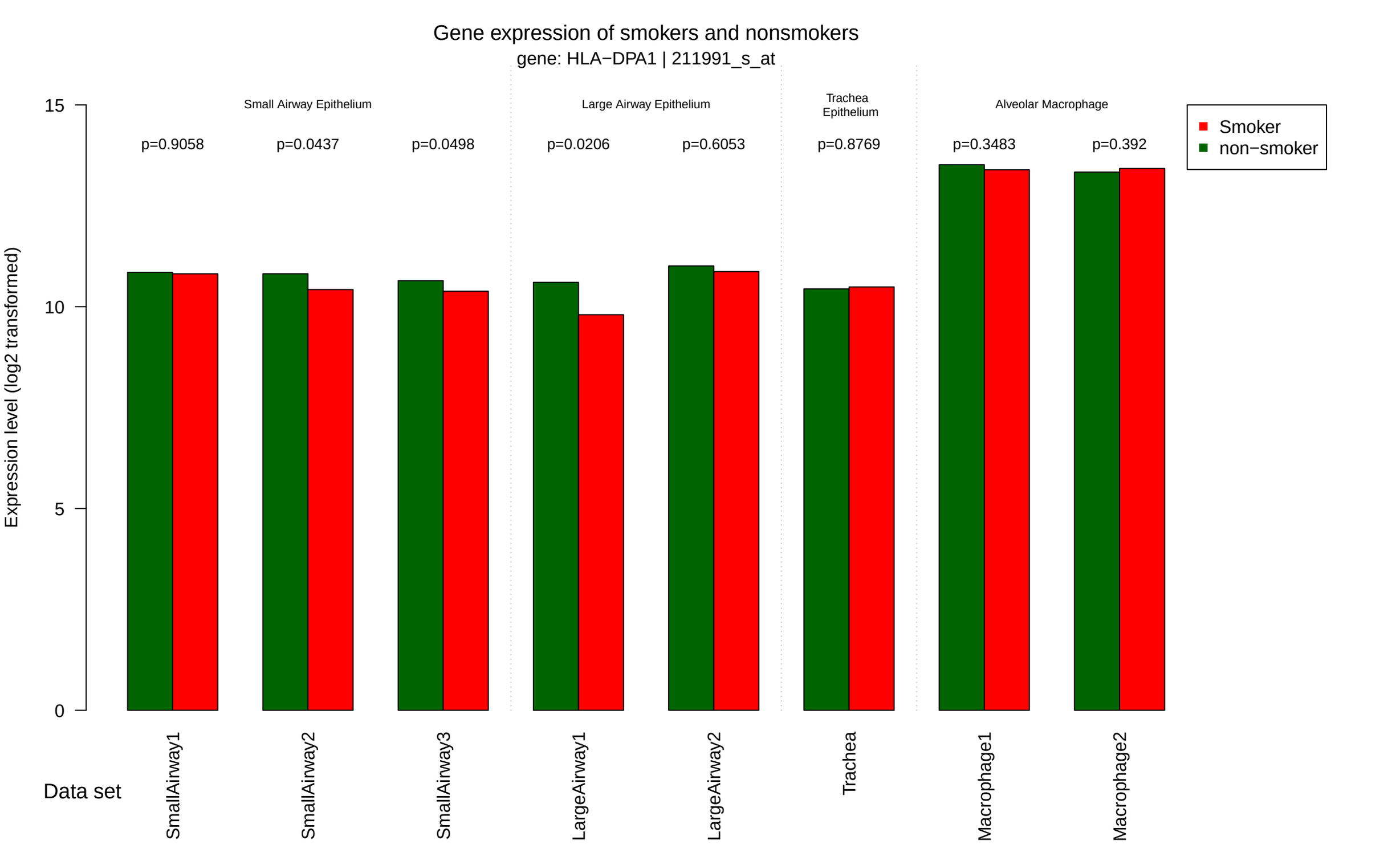
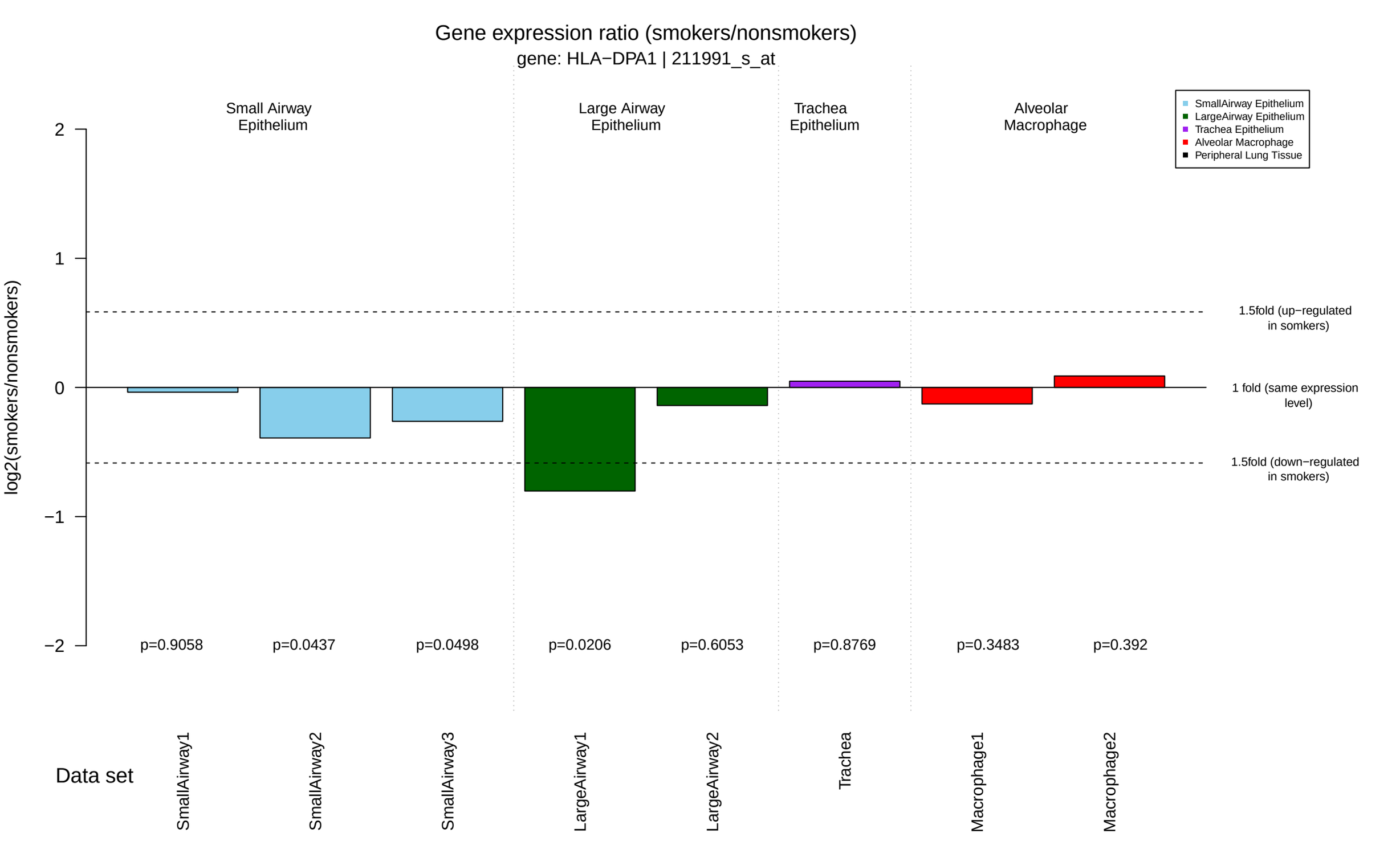
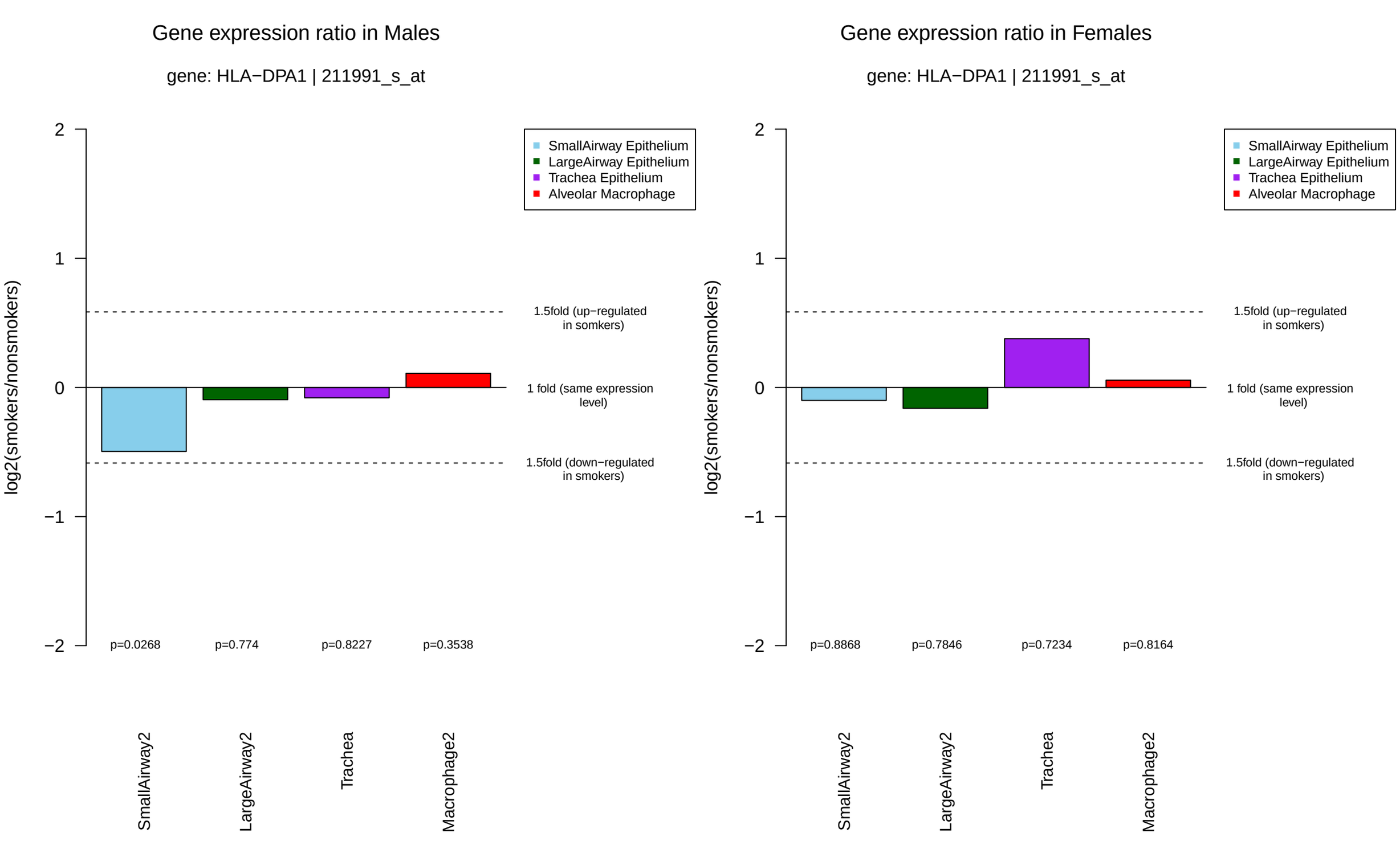
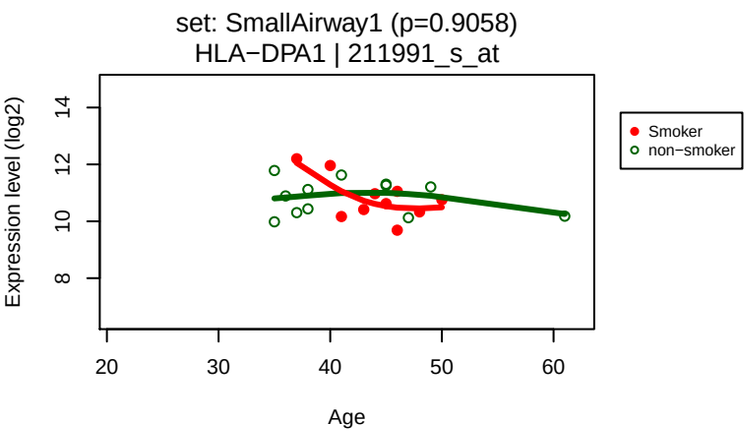
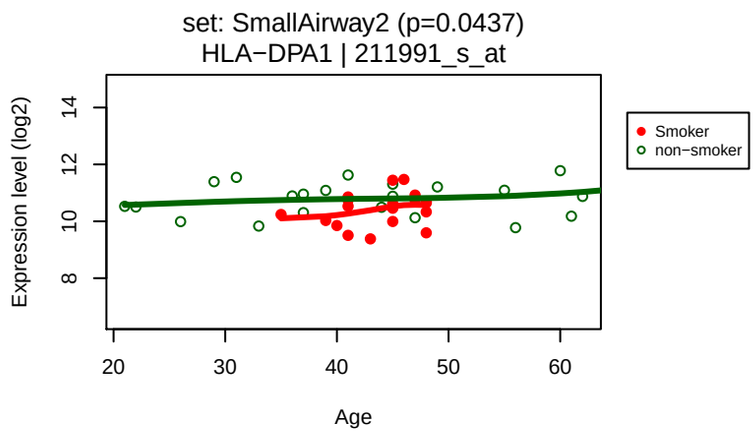
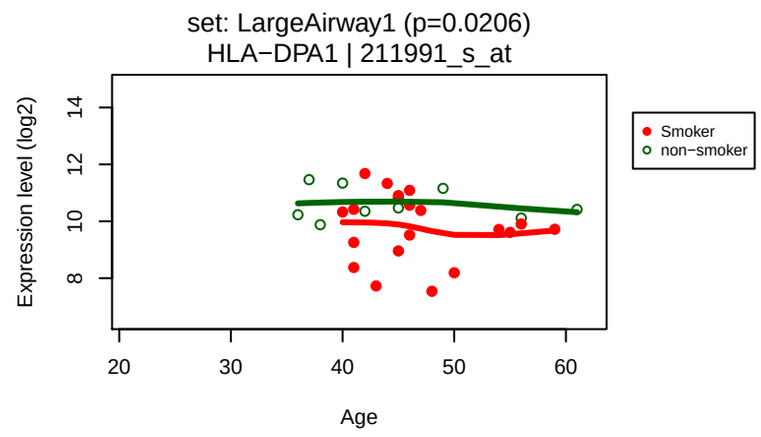

Gene expression of smokers and nonsmokers
gene: HLA−DPA1 | 211991_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9058
p=0.0437
p=0.0498
p=0.0206
p=0.6053
p=0.8769
p=0.3483
p=0.392
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

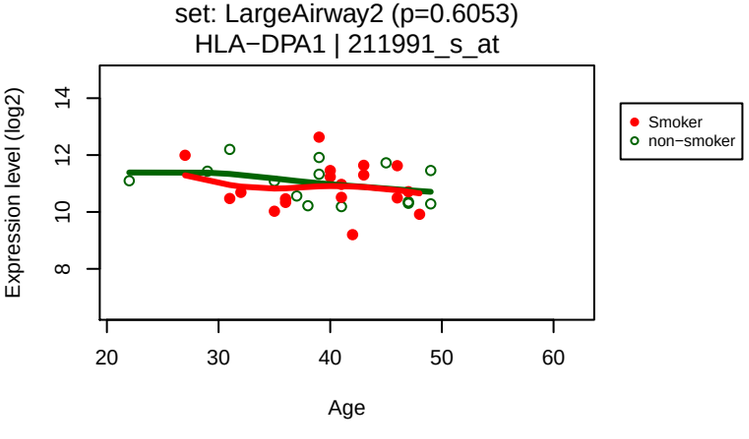

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DPA1 | 211991_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9058
p=0.0437
p=0.0498
p=0.0206
p=0.6053
p=0.8769
p=0.3483
p=0.392
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

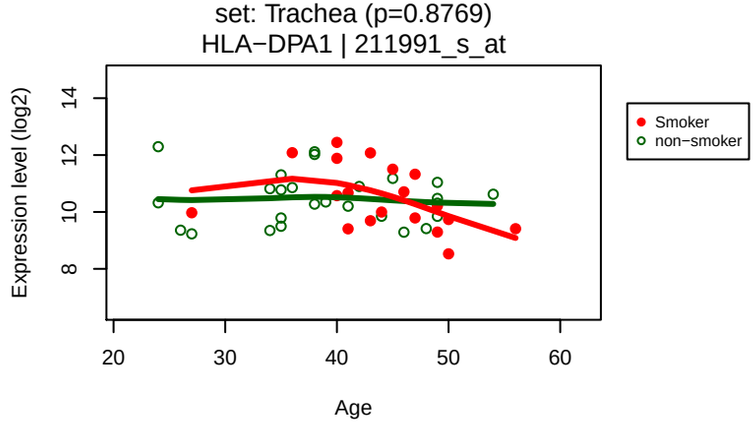

Gene expression ratio in Males
gene: HLA−DPA1 | 211991_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0268
p=0.774
p=0.8227
p=0.3538
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DPA1 | 211991_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8868
p=0.7846
p=0.7234
p=0.8164
SmallAirway2
LargeAirway2
Trachea
Macrophage2

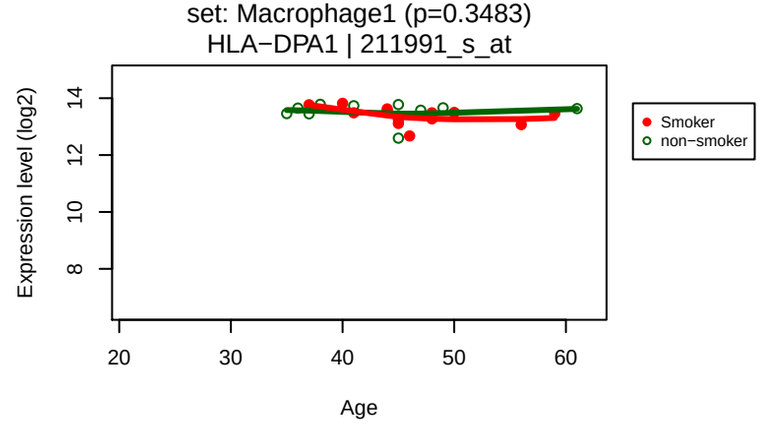

set: SmallAirway1 (p=0.9058)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker

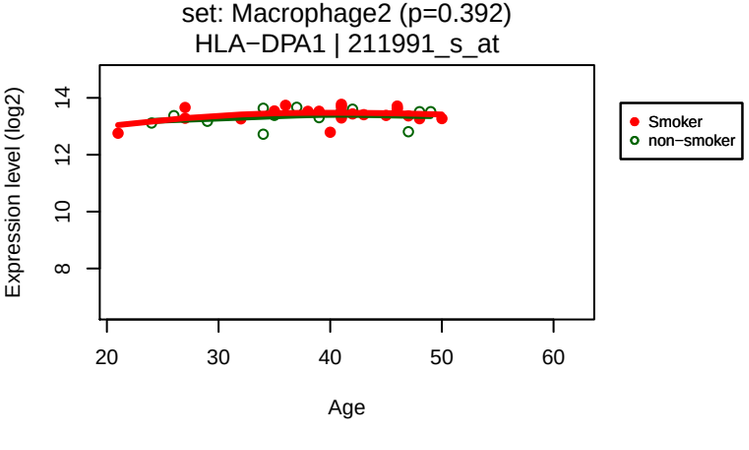

set: SmallAirway2 (p=0.0437)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker

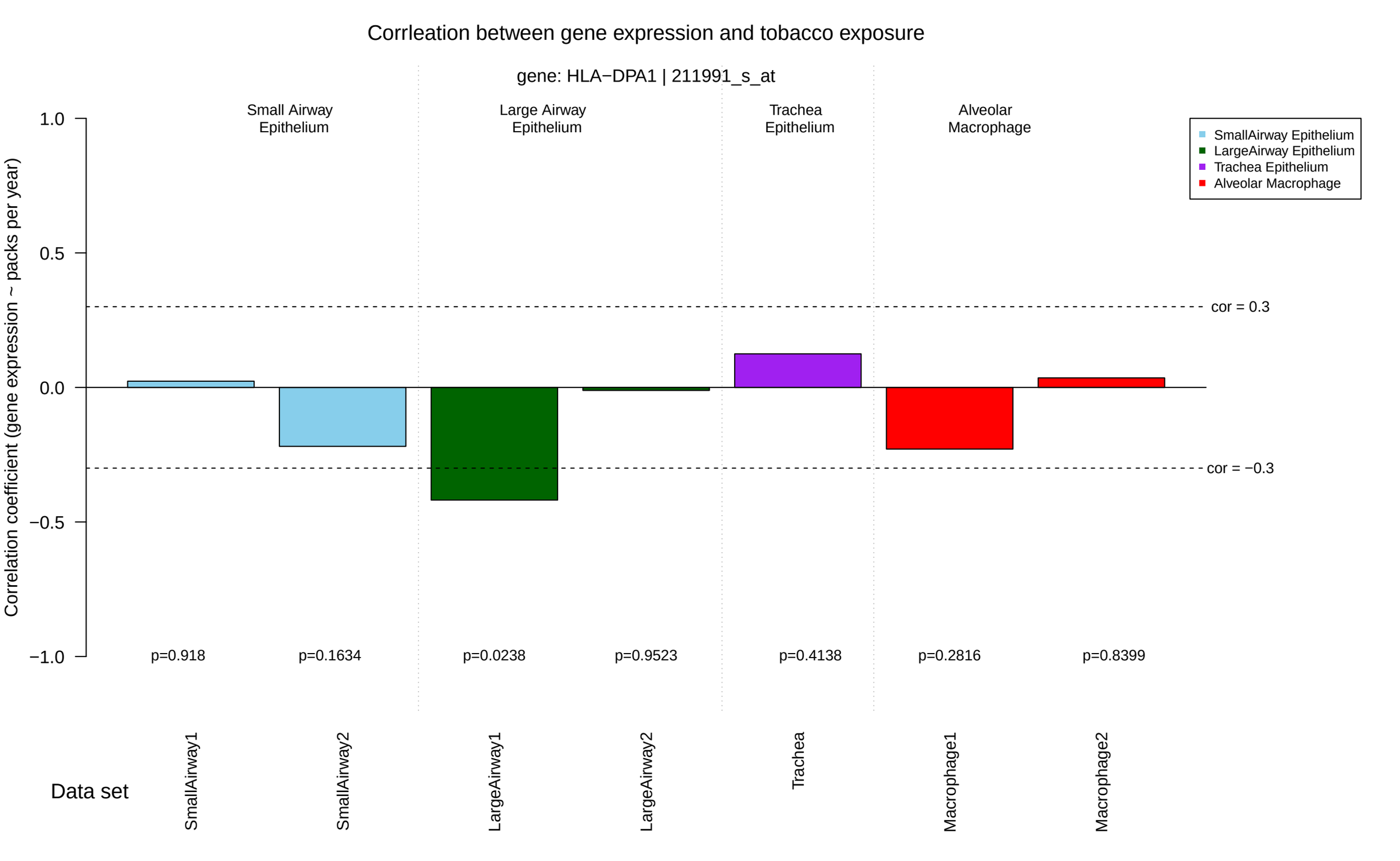

set: LargeAirway1 (p=0.0206)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6053)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8769)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3483)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.392)
HLA−DPA1 | 211991_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DPA1 | 211991_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.918
p=0.1634
p=0.0238
p=0.9523
p=0.4138
p=0.2816
p=0.8399
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2